

A Width-Gap Criterion for Compact Images

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Source and Target Problem

This packet concerns Problem 8.1 of Oikhberg and Ostrovskii [1], arXiv:1205.0095, “Dependence of Kolmogorov widths on the ambient space.” The source asks:

Let K be a compact in a Banach space X and $T : X \rightarrow Y$ be a compact operator. Does it follow that $d_n(TK) = o(d_n(K))$?

The purpose of this section is to make some comments on the following intriguing problem

Problem 8.1. *Let K be a compact in a Banach space $\mathcal{X}$ and $T : \mathcal{X} \rightarrow \mathcal{Y}$ be a compact operator. Does it follow that $d_n(TK) = o(d_n(K))$?*

Set $\hat{d}_n(K) = d_n(K, \overline{\text{span}[K]})$. [OS09, Lemma 6.1] states:

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The result below is a partial positive criterion for this problem. It does not settle the full problem.

Definitions

For a bounded set A in a Banach space Z , let

$$d_n(A, Z) = \inf_{\dim E \leq n} \sup_{a \in A} \text{dist}(a, E)$$

denote the Kolmogorov n -width. For a bounded operator $T : X \rightarrow Y$, set

$$\beta_r(T) = d_r(TB_X, Y), \quad r = 0, 1, 2, \dots$$

If T is compact, then TB_X is relatively compact, hence $\beta_r(T) \rightarrow 0$.

Idea of the Proof

The proof spends the available n dimensions in two accounts. Use $n - r$ dimensions to approximate K inside X . The error vector then lies in a small multiple of the unit ball of X , and applying T sends that error into the compact set TB_X . Spend the remaining r dimensions to approximate TB_X . This gives a direct inequality with a tunable parameter r . Proposition 8.3 in the source corresponds to keeping r fixed after choosing it so that $\beta_r(T)$ is small; the criterion here also allows r to vary with n .

Main Result

Theorem 1 (Width-gap inequality). *Let X, Y be Banach spaces, let $T : X \rightarrow Y$ be bounded, and let $K \subset X$ be bounded. For every $0 \leq r \leq n$,*

$$d_n(TK, Y) \leq \beta_r(T) d_{n-r}(K, X).$$

Consequently,

$$d_n(TK, Y) \leq \inf_{0 \leq r \leq n} \beta_r(T) d_{n-r}(K, X).$$

Proof. Fix $0 \leq r \leq n$ and $\varepsilon > 0$. Choose a subspace $F \subset Y$ with $\dim F \leq r$ such that

$$TB_X \subset F + (\beta_r(T) + \varepsilon)B_Y.$$

Choose a subspace $E \subset X$ with $\dim E \leq n - r$ such that

$$\sup_{x \in K} \text{dist}(x, E) \leq d_{n-r}(K, X) + \varepsilon.$$

Set $G = TE + F$. Then $\dim G \leq n$.

For $x \in K$, choose $e \in E$ with $\|x - e\| \leq d_{n-r}(K, X) + 2\varepsilon$. Since

$$Tx = Te + T(x - e)$$

and $T(x - e) \in \|x - e\| TB_X$, the choice of F gives

$$\text{dist}(Tx, G) \leq (\beta_r(T) + \varepsilon)(d_{n-r}(K, X) + 2\varepsilon).$$

Taking the supremum over $x \in K$, then the infimum over admissible G , and finally letting $\varepsilon \downarrow 0$, proves the claim. $\square$

Corollary 1 (Operator-adapted positive cases). *Let $K \subset X$ be compact and let $T : X \rightarrow Y$ be compact. If*

$$\inf_{0 \leq r \leq n} \frac{\beta_r(T) d_{n-r}(K, X)}{d_n(K, X)} \rightarrow 0$$

along all n for which $d_n(K, X) > 0$, then

$$d_n(TK, Y) = o(d_n(K, X)).$$

In particular, the conclusion holds whenever for every $C > 1$ and each fixed r one eventually has

$$d_{n-r}(K, X) \leq C d_n(K, X),$$

which recovers Proposition 8.3 of [1].

Proof. The first assertion follows immediately from Theorem 1. For the final statement, fix $\delta > 0$. Since T is compact, choose r with $\beta_r(T) < \delta$. The finite-shift hypothesis gives, for all sufficiently large n ,

$$d_n(TK, Y) \leq \beta_r(T)d_{n-r}(K, X) \leq C\delta d_n(K, X).$$

Letting $\delta \downarrow 0$ proves the claim. $\square$

Verification and Scope

The proof is purely functional analytic. It uses only the definition of Kolmogorov width and the compactness fact $\beta_r(T) \rightarrow 0$. The crop in ‘figures/open_problem_crop.png’ was generated from page 18 of the source PDF by ‘code/crop_open_problem.py’.

This result is a partial theorem, not a full solution of Problem 8.1. The unresolved case is exactly where the width sequence of K has large enough finite gaps to defeat every possible choice of r in the criterion.

Bounded novelty check: local run indexes were searched for arXiv 1205.0095, Kolmogorov widths, ambient space, and compact-operator width keywords, with no duplicate packet found. A web search for the exact phrases “ $d_n(TK) = o(d_n(K))$ ”, “Widths of images of compacts under compact operators”, and variants returned only the source paper or irrelevant width papers, not a later answer to Problem 8.1.

Human Review Recommendation

Review Theorem 1 first. If accepted, the packet should be recorded as a likely valid partial result for Problem 8.1: it gives a compactness-rate criterion and clarifies that Proposition 8.3 is a fixed finite-shift special case.

References

- [1] T. Oikhberg and M. I. Ostrovskii, *Dependence of Kolmogorov widths on the ambient space*, arXiv:1205.0095, 2012.
- [2] M. I. Ostrovskii and V. S. Shulman, *Weak operator topology, operator ranges and operator equations via Kolmogorov widths*, Integral Equations Operator Theory 65 (2009), 551–572.